

SSQ-LR: Sub-FP16 Recurrent State Quantization for Hybrid Reasoners

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Abstract

This is a living paper shell for a preregistered Mac-first experiment. SSQ-LR tests whether recurrent SSM state in hybrid reasoning models can be quantized below FP16 while preserving quality. No positive result or native systems claim is made yet.

1 Current Gate

The branch must pass state-distribution heterogeneity, state-only quantization sensitivity, and cross-model transfer gates before any GPU validation. S1 must first show real state heterogeneity on a live hybrid model; S2 then freezes one state-only quantization recipe; S3 tests that recipe without per-model retuning. Failing any gate kills the branch before GPU work.

2 Mac Artifact Status

A deterministic synthetic S1 packet now validates the real-schema checker path before real hybrid-model state dumps. It emits 288 rows covering 12 prompts, 6 recurrent layers, and the four preregistered buckets, producing `SCHEMA_REHEARSAL_NOT_PROMOTABLE_SYNTHETIC_SSQ_LR_S1`. This is synthetic-only: it is not model evidence, not a quality result, and not GPU evidence. The metadata-only model eligibility packet identifies live hybrid targets (granite-4.0-h-tiny, granite-4.0-h-small, granite-4.0-h-small-FP8, and Qwen3-Next-80B-A3B-Instruct) and ties them to shared architecture-map hashes, but all remain blocked for real S1 on this Mac because no corresponding model weights are cached repo-locally.

3 Next Evidence

The next admissible row is a real SSM-state dump from the smallest available hybrid reasoner. Promotion requires S1 heterogeneity on real states before any state-quantization recipe is tuned.

4 Admissible Real Packet

The real S1 packet must include model and tokenizer revisions, prompt manifest hash, prompt_ids_hash, seed list, context lengths, dtype, device, command, and the architecture_map_hash from the shared config-derived maps plus the trace_plan_hash from the frozen SSQ-LR trace plan. Promotable packets must also include the exact trace_plan_path; its file SHA-256 must match the registered trace_plan_hash, so caller-created subsets are resource-limited readouts rather than S1 promotion evidence. The current capture-manifest templates are generated by `experimental.shared.hybrid.trace.capture.manifest`; they must be filled from real SSM-state captures before packet building. Served HF IDs are preserved as `served_model_id` but rows are validated against the canonical architecture-map `model_id`. Rows must report `model_id`, `model_revision`, `prompt_id`, `layer`, `position_bucket`, `state_shape`, `max_abs`,

rms, std, kurtosis, outlier_mass, and a bf16_no_quant control. The packet is valid only if every prompt/layer pair covers prefill_end, 2k_or_end, 8k_or_end, and final_minus_128 position buckets and at least 12 prompt IDs unless config.json records an explicit resource-limit note. Rows must identify an SSM/Mamba layer_kind and recurrent state_tensor_kind, so arbitrary activation tensors cannot promote S1. Any resource-limited packet must use a RESOURCE_LIMITED_NOT_PROMOTABLE decision. A schema rehearsal must set schema_rehearsal: true and use a SCHEMA_REHEARSAL_NOT_PROMOTABLE decision. The summary must include the recomputed S1 evaluator fields: gate_name/status/pass flag, prompt count, position buckets, SSM layer count, required and observed passing-layer counts, distribution-passing layer counts, selected S1 ratio and prompt-level lower bound, Holm-adjusted two-sample readout, magnitude/distribution pass flags, a 1.25x distribution-effect floor flag, and final-minus-128 versus prefill-end max-abs/std/kurtosis ratios. The checker recomputes these fields from raw rows rather than global layer means and rejects stale summaries; a tiny distribution-only shift cannot promote S1, and a global mean spike in one layer cannot satisfy the distribution path unless the counted layer/metric tests also clear the per-layer 1.25x effect-size floor. It must pass experimental.shared.check_gate_packet --mode real --project ssq_lr. Saved tensor packets must be converted with experimental.shared.hybrid_trace_packet_builder before validation. The converted packet must include tensors/tensor_manifest.json and the referenced tensor artifacts; each raw row copies tensor_name, tensor_source_name, tensor_storage_name, tensor_sha256, tensor_dtype, and tensor_shape from that manifest, and state_shape must match the saved tensor_shape.

Required baseline	Purpose
BF16 no-op replay	Catch hook/serialization bugs.
INT8 state-only	Standard low-bit state control.
FP8-style state-only	Checks whether sub-FP16 state can match the likely deployed fallback.
MXFP4 state-only	Candidate sub-FP16 recipe.
Random same-L2 noise	Separate structure from noise magnitude.
Scale shuffle	Test scale assignment sensitivity.
Byte accounting	Include scale/metadata overhead.

Table 1: Controls required before SSQ-LR can claim a state-quantization recipe.

5 Reviewer Packet

The reviewer-facing claim boundary is tracked in experimental/ssq_lr/paper/reviewer_pack.md. It records that SSQ-LR is not camera-ready as a method paper until real S1–S3 evidence exists.

6 Claim Boundary

SSQ-LR is only about recurrent SSM-state precision in hybrid reasoners. It does not claim weight quantization, activation quantization, KV-cache quantization, native serving speed, or memory savings until a real state recipe and later GPU validation pass their preregistered gates.